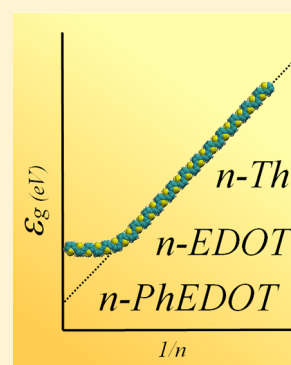


Reviewing Extrapolation Procedures of the Electronic Properties on the π -Conjugated Polymer LimitJuan Torras,^{*,†} Jordi Casanovas,[‡] and Carlos Alemán^{§,||}[†]Departament d'Enginyeria Química, Escola d'Enginyeria d'Igualada, Universitat Politècnica de Catalunya, Pça Rei 15, Igualada 08700, Spain[‡]Departament de Química, Escola Politècnica Superior, Universitat de Lleida, c/Jaume II no. 69, Lleida E-25001, Spain[§]Departament d'Enginyeria Química, E. T. S. d'Enginyeria Industrial de Barcelona, Universitat Politècnica de Catalunya, Diagonal 647, Barcelona E-08028, Spain^{||}Centre for Research in Nano-Engineering, Universitat Politècnica de Catalunya, Campus Sud, Edifici C', C/Pasqual i Vila s/n, Barcelona E-08028, Spain

Supporting Information

ABSTRACT: In this article, the extrapolation procedures of π - π^* electronic transition energy on π -conjugated oligomers are reexamined. Different models, including the simplest coupled oscillator, the free electron, the Hückel approach, the molecular exciton model, and some specific fitting-functions, are compared using the transition energies derived from theoretical calculations on three thiophene-based oligomer series. Specifically, oligomers of up to 30 repeating units have been considered to include the saturation effects as a function of chain length. The coupled oscillator model of W. Kuhn and the fitting-function of Hirayama are the models that present the better suit on the transition energy interpolation as a function of chain length. Using only the first four oligomers of the series ($n = 2$ up to 8) yields an estimation of the transition energy on the polymer limit with an average error of $\sim 1.5\%$. The vertical and adiabatic ionization potential present a better fit with the Hückel model approach. Finally, implications of the environmental polarity on the electronic properties, molecular geometry, charge distribution, and aromaticity are shortly discussed.



INTRODUCTION

Conjugated organic polymers, presenting a semiconducting or conducting behavior, have received considerable interest because of their wide range of technological applications in fields such as electronics, biomedical engineering, and optics.¹ The alternating pattern combining carbon-carbon single and double bonds, lying on the conjugated polymer backbone, provides a π -orbital overlap along the molecule. The structural bonds arrangement on π -conjugated polymers (π -CPs) and oligomers (π -COs) gives a number of useful properties such as easy charge transport and band gap modeling.²

Among π -CPs and π -COs, those based on thiophene (Th) present a high environmental stability of their doped and undoped states, as well as an easy way to tune their chemical structure to control their electronic properties.¹ Accordingly, in order to understand structural and environmental effects on the band gap of those compounds, an increasing number of experimental and theoretical works have been conducted.¹

However, the oxidation of π -CPs (p-doping) is typically associated with a shift in the absorption energy π - π^* , resulting in an observable color change. The study of the absorption peaks evolution becomes crucial for understanding electrochromic properties and for characterizing the charge carriers produced during doping.³ Geometry studies concerning neutral and oxidized forms of π -CPs show a distortion from the

benzenoid-type structure of the former to the quinoid-type structure on the latter.³ Thus, the bond length alternation (BLA), defined as the difference between the average length of C-C double and single bonds, represents the largest contribution to the energy band gap (ϵ_g).⁴ Consequently, structural modifications on the π -CPs that leads to a change in the BLA, in turn, can produce a decrease in the π - π^* transition energy.⁵

An extended and generally accepted way to obtain electronic information data from oligomer calculations is by plotting the results against inverse chain length and extrapolating to infinity. This approach is well established for ϵ_g and ionization potentials (IPs). Usually, the agreement and the correlation coefficients between predicted and observed ϵ_g values are very good, if the short-chain oligomers are excluded.⁶ Within this context, the linear extrapolation of the $E_{\text{HOMO}}-E_{\text{LUMO}}$ gap obtained for π -COs against the reciprocal of the number of repeating units ($1/n$) has been generally considered as an acceptable procedure to predict the ϵ_g for the corresponding π -CPs.

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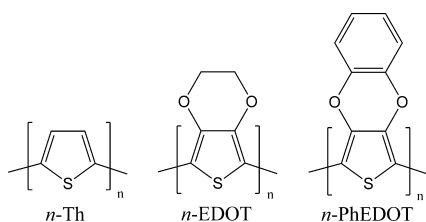
Nonlinear optical (NLO) response in quasi-one-dimensional systems was an active area of research in the nineties⁷ when the relationship between the π -electron delocalization and the second hyperpolarizability (γ) of π -CPs was established. These systems present very large electron–phonon coupling, vibrational displacements being strictly related to electronic excitations.⁸ Consequently, electron correlation leads to a diminution of the BLA parameter⁹ and the subsequent saturation of both the γ ¹⁰ and the optical transition energies with increasing chain length.¹¹

Studies related with the saturation of different electronic properties [e.g., ϵ_g , the molecular polarizability (α), and γ] were extended to heterocyclic systems, looking for functions best suited to describe this saturation.¹² Also, the solvent's influence was considered to explain the discrepancy between different points of saturation observed when comparing theoretical and experimental results for a given π -CP.¹³ Indeed, theoretical models usually refer to idealized planar structures of π -CPs, whereas experimental measurements are typically associated with some conformational disorder that would reduce the extent of electron delocalization. The connection between chain length, conjugation length, and electronic properties saturation in chains with conformational disorder has been pointed out on the bibliography¹⁰ and treated more explicitly by the molecular exciton formalism.¹⁴

Experimental results showed that extrapolation fails to consider an asymptotic behavior to derive the electronic properties from the linearization with the inverse of the chain length, thus became more evident the need to include the saturation effects in their extrapolation to infinite chain.¹¹ In fact, Meier et al.¹⁵ pointed out that saturation of the electronic properties of π -CPs occurs around 12 repeating units. They defined the effective conjugation length (ECL) as the conjugation length at which the wavelength of the absorption maximum in a set of oligomers with increasing number of repeating units is not more than 1 nm above the lower limit, which is given by the infinitely long polymer chain (i.e., $\lambda_\infty - \lambda_{\text{ECL}} \leq 1$ nm). More recently, different authors suggested that saturation effects should be considered in the extrapolation of these electronic properties have with the chain length, different approximations being proposed.^{11,16}

In our understanding, there is a lack of overview on the different approaches used so far to obtain the most suited extrapolation in the electronic properties of π -COs systems at the infinite chain. In this article, we propose a review of these approaches from a practical point of view considering π -CPs of three different thiophene-derivatives (Scheme 1) and taking into account their saturation effect. Specifically, studies concerning to the saturation of the electronic properties (i.e., ϵ_g , IP, BLA, atomic charge, spin density, and aromaticity) in environments of growing polarity have been performed.

Scheme 1



The rest of the article is organized as follows: first, we review the most used extrapolating methodologies at infinite chain in order to obtain the electronic properties from both experimental and theoretical data of π -COs. Then, we overview some of the most representative extrapolation methods being applied to three different thiophene-derivative oligomers. We then extract some indicators in order to analyze their performance considering the saturation effect within a given model chemistry using the density functional theory (DFT) approach. Finally, we discuss the effect of the environment polarity on the electronic properties of the analyzed systems.

MATERIALS AND METHODS

Theoretical Background. A procedure widely used to get an estimation of the electronic properties at the infinitum polymer limit is the linear extrapolation. Thus, either values of π - π^* transition or HOMO–LUMO gap are usually plotted as a function of $1/N$ or $1/n$, where N and n are the number of π -electrons and repeating units contained in each oligomer, respectively. The linearity of this representation is maintained for the first elements of the series (up to $n \approx 12$), which stimulates many groups to use that approach in order to obtain the electronic property limit (i.e., for N or $n \rightarrow \infty$) using the smaller oligomers only. However, both experimental and theoretical values suffer from saturation to levels above $n \approx 12$ leading to a defect-estimation of ϵ_g when carrying out such linearizations (Figure 1). Several approximations have been

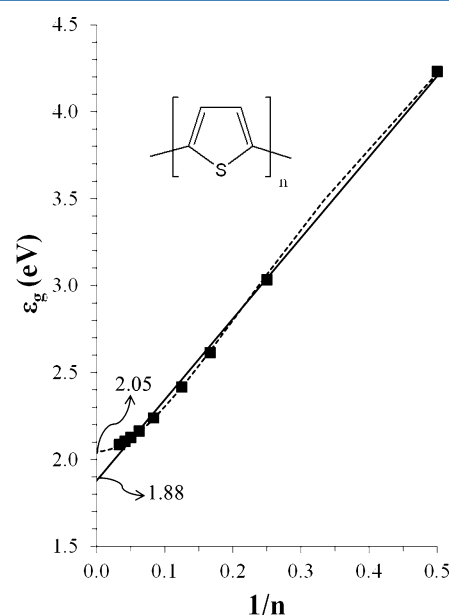


Figure 1. Theoretical HOMO–LUMO gap (ϵ_g) of oligothiophenes n -Th in gas phase against the inverse of the chain length. Extrapolations to infinite chain length (ϵ_g^∞) have been performed using the W. Kuhn fit (eq 8) and the inverse of the chain length fit (eq 3a), which are represented by dashed and solid lines, respectively.

proposed to get reliable estimated values of ϵ_g for the infinitum polymer limit, which will be discussed in the following sections of this work.

Free-Electron Model. An early model used to explain the behavior of π -COs is the free-electron molecular orbital (FEMO), in which the whole molecule structure is considered as one-dimensional potential box with an infinite barrier height.

The potential inside the box is set as zero, whereas an infinity value is considered outside the box. The energy levels are

$$E_N = \frac{h^2}{8mL^2} N^2 \quad (1)$$

where N is the number of π -electrons; h is the Planck constant; m is the mass of the electron; and L is the width of the box. In order to simplify, only π -COs with an even number of π -electrons are considered.¹⁷ Under this consideration, if the average bond length between two carbon atoms is named d_{C-C} , the whole skeleton of a conjugate molecule will be $L = (N - 1)d_{C-C}$. The ϵ_g between the highest occupied level ($N/2$) and the lowest free level ($1 + N/2$) energies from the FEMO model is easily obtained using eq 1:

$$\epsilon_g = E_{(1+N/2)} - E_{(N/2)} = \frac{h^2}{8md_{C-C}^2} \frac{N + 1}{(N - 1)^2} \quad (2)$$

By finding a power-series expansion of the eq 2 in powers of $1/N$, we obtain:¹⁸

$$\epsilon_g = \frac{h^2}{8md_{C-C}^2} \left[\frac{1}{N} + \frac{3}{N^2} + O(N^{-3}) \right] \quad (3)$$

This simple power-series expansion can easily explain the so often used empirical equation that describes the ϵ_g of the long-wavelength transition as a linear function of $1/N$ or $1/n$ (eq 3a). Also, the deviation from linearity could be explained when considering the quadratic term (eq 3b).

$$\epsilon_g = a + b \frac{1}{N} \quad (3a)$$

$$\epsilon_g = a + b \frac{1}{N} + c \frac{1}{N^2} \quad (3b)$$

However, an approximated model, named *extended FEMO*, was formulated by H. Kuhn when in a seminal paper introduced a one-dimensional periodical potential inside the quantum box where free-electrons are delocalized.¹⁷ Concretely, considering the unsymmetrical distance between single and double bonds on the π -CO skeleton of C–C bonds, a sine potential is introduced inside the quantum box. Also, there is coincidence of the crests and troughs of the sine wave with the longer and shorter bonds, respectively. The BLA between longer and shorter C–C distances brings about a disturbance in the behavior of the π -electron gas along the chain that will be reflected on the wavelength of the first absorption band. Accordingly, H. Kuhn proposed the following approximation for the ϵ_g on the extended FEMO model:

$$\epsilon_g = \frac{h^2}{8mL^2} (N + 1) + V_0 \left(1 - \frac{1}{N} \right) \quad (4)$$

where L is the length of the conjugated chain (along the zigzag line), N is the number of π -electrons, and V_0 is the amplitude of the sine curve used to approximate the variations of the potential energy between single and double bonds. The extended FEMO model has been successfully applied to polyenes. However, Th-containing oligomers, n -Th, can be considered as polyenes with a fixed skeleton due to the rigidity of the Th ring and the fact that the inter-ring bond contributes to the overall structure. Colditz et al.¹⁹ applied eq 4 to several spectroscopic properties of the n -Th, with n up to 6. They considered a box length $L = (N+1)d_{C-C}$, where $N = 4n$.

However, this approximation does not take into account that the effective length of the conjugated chain is not just the distance between the two terminal carbon atoms. Taubmann¹⁸ proposed to consider the shape of the π -orbitals in the definition of the conjugated skeleton (i.e., specifically the π^* orbitals, which stretch out beyond the end of the C=C bond). Thus, the effective length of the box (L) is longer than the distance between the two terminal carbon atoms. Therefore, the effective length L was proposed to be estimated by using a fitting parameter l : $L = (N + l)d_{C-C}$. Under this scheme, π -electrons are allowed to move a distance ld_{C-C} beyond the terminal carbon atoms.¹⁸ In fact, on the basis of the Taubmann consideration for the effective length of the conjugated molecule on the extended FEMO theory, Seixas de Melo et al.²⁰ successfully applied a three fitting parameters equation (eq 5) to suit spectroscopic data of n -Th, with n up to 7. In this approximation, an infinite band gap (ϵ_g^∞) of 2.0–2.1 eV was obtained for the PTh singlet state, reproducing the experimental value of 2.1 eV obtained directly from absorption optical data.²¹

$$\epsilon_g = \frac{h^2}{8md_{C-C}^2} \frac{(N + 1)}{(N + l)^2} + V_0 \left(1 - \frac{1}{N} \right) \quad (5)$$

Moreover, a very close HOMO–LUMO transition gap limit (2.03 eV) was obtained by applying a quadratic polynomial extrapolation of ϵ_g against $1/n$ to the results derived from theoretical calculations on n -Th oligomers with n up to 50.^{16b}

Coupled Oscillators Model. In a seminal paper based on classical mechanics, Lewis and Calvin²² proposed that formal double bonds of a polyene can be split into individual units of double bonds (C=C) representing N identical oscillators with a specific vibrational energy (i.e., end effects are ignored in this approach). The oscillation frequency of an oscillator with mass $N\mu_0$ and restoring force constant k_0 can be expressed as

$$\nu_0 = \frac{1}{2\pi} \sqrt{\frac{k_0}{N\mu_0}} \quad (6)$$

By expressing the frequency as a wavelength λ and using a new constant, which includes the effective mass of one unit, the following equation is easily obtained:

$$\lambda^2 = k_0' N \quad (7)$$

Equation 7 was verified on the diphenylpolyenes series of increasing chain length, λ^2 of the first absorption maxima being plotted against N . Results showed of the linear behavior of λ^2 for the higher values of N .²²

Later on, W. Kuhn²³ used the work of Lewis and Calvin (eq 7) to propose the inclusion of a coupling force constant, k' , between contiguous oscillators units

$$\nu = \nu_0 \sqrt{1 + 2 \frac{k'}{k_0} \cos \left(\frac{\pi}{N + 1} \right)} \quad (8)$$

where $\nu_0 = (k_0/4\pi^2\mu_0)^{1/2}$ is the frequency of an isolated oscillator unit with $k'/k_0 \approx -0.46$. Interestingly, adjustment of eq 8 to polyphene data using the number of repeating units, instead of the number of double bonds, results in $k'/k_0 \approx -0.32$. A simple linearization of the ϵ_g^∞ can be obtained from the W. Kuhn approximation by representing the squares of the band gap against the cosine of $\pi/(N+1)$.

Hückel Model. A very simple model to deal with π -COs systems is the Hückel molecular orbital (HMO) theory.

Proposed in the early thirties,²⁴ it consists on a simple linear combination of atomic orbitals of molecular orbitals (LCAO-MO) involving π -electrons. This theory easily allows us to predict the MO energy levels

$$E_k = \alpha + 2\beta \cos\left(\frac{k\pi}{N+1}\right) \quad k = 1, 2, \dots, N \quad (9)$$

where N is the number of conjugated carbon atoms (π -electrons) and both α and β are empirical parameters usually estimated from experimental energies. However special attention should be paid to the method of estimation since β governs the spacing of the MO energy levels, and different values of β can be obtained depending on the method of estimation.¹⁸ An interesting review describing the historical background and some special applications of the HMO model has recently been published.²⁵ This review highlights historical background and some other features of the HMO model, such as the closed solutions for some special classes of conjugated hydrocarbons. In the case of polyenes with an even number N of π -electrons, the HOMO–LUMO gap is easily obtained using eq 9:

$$\begin{aligned} \varepsilon_g &= 2\beta \left\{ \cos\left[\frac{\pi}{2}\left(1 + \frac{1}{N+1}\right)\right] - \cos\left[\frac{\pi}{2}\left(1 - \frac{1}{N+1}\right)\right] \right\} \\ &= -4\beta \sin\left[\frac{\pi}{2}\left(\frac{1}{N+1}\right)\right] \end{aligned} \quad (10)$$

Interestingly, despite that the FEMO model seems to be simpler than the HMO, since it deals only with kinetic energy changes and ignores the potential energy, Ruedenberg et al.²⁶ demonstrated that such two models are essentially equivalent. This equivalency elucidates the role played by the length of the bond paths in the semiempirical HMO energy parameters and permits to visualize the LCAO coefficients as amplitudes of harmonic network waves at the atomic positions on the bond paths.²⁵ Indeed, in order to determine the β parameter by mixing with the FEMO theory, Taubmann¹⁸ took advantage of this equivalency and of the electronic properties limit at the infinite polyene chain. Consequently, after expanding eq 10 in a Taylor series and being ordered in powers of N^{-1} , ε_g was expressed as

$$\varepsilon_g = -2\pi\beta \left[\frac{1}{N} - \frac{1}{N^2} + O(N^{-3}) \right] \quad (11)$$

It can be seen from eqs 3 and 11 that both $\varepsilon_g^{\text{FEMO}}$ and $\varepsilon_g^{\text{HMO}}$ have a quadratic behavior with $1/N$ as N tends to infinity. As both $\varepsilon_g^{\text{FEMO}}$ and $\varepsilon_g^{\text{HMO}}$ energies are expressions of the HOMO–LUMO energy difference from two different models, a relationship between $d_{\text{C-C}}$ and β can be obtained by matching the coefficients of N^{-1} in eqs 3 and 11.¹⁸

More recently, Materny et al.²⁷ used a LCAO-MO model based on a transition from the HMO to a tight-binding approach²⁸ to derive a chain-length-dependent expression for the ε_g from absorption data of stable diacetylene oligomers (n -DA). Thus, by taking into account the changes in the distances between carbon atoms due to the alternation of σ and π bonds on the poly(diacetylene) (PDA) chain, the HOMO–LUMO transition energy was evaluated by

$$\varepsilon_g = 2\sqrt{\beta_1^2 + \beta_2^2 + 2\beta_1\beta_2 \cos\left(\frac{2n\pi}{2n+c}\right)} \quad (12)$$

where β_s are the interaction energies between two adjacent p and π orbitals (resonance integrals) for each one of the involved bonds, and c is a factor to correct the expression for chains with finite length. The parameters β_1 , β_2 , and c are typically obtained by fitting to experimental data. Notice the similarity in the expressions provided in eq 12 (HMO theory) and eq 8 (coupled oscillator theory) to model π -COs. Similar expressions were found on structures of finite and infinite linear, cyclic, and Möbius strip polyacenes by adopting the Hückel and MNDO schemes.²⁹ Interestingly, the Möbius belting process led to an evolution of ε_g with the chain length similar to the linear polyacenes rather than their cyclic analogues. Also, another interesting method to treat for quasi-one-dimensional polymeric systems was addressed by Cui et al.,³⁰ who exploited the relationship in HMO theory between the energy levels and orbitals of an infinite periodic chain and those of a finite chain. More recently, this method has been extrapolated to Hartree–Fock and Kohn–Sham DFT calculations on rather long oligomers.³¹

Following with similarities and equivalences between the HMO and other proposed models on the bibliography, it is worth noting that the study of Onipko et al.,³² which was based on the Hückel Hamiltonian, showed an asymptotic behavior of the HOMO–LUMO gap for π -COs of type $M-M_{n-2}-M$ and $M-M_n-M_1$ with $M = M_1-M_2$, where M , M_1 , and M_2 are arbitrary monomers. Then, a simple linear expression was suggested to represent ε_g as a function of oligomer length.

$$\varepsilon_g = \varepsilon_g^\infty + k\frac{1}{n^2} \quad (13)$$

Molecular Excitons Theory. An initial description of the excitation theory was first proposed by Frenkel in 1931.³³ This author showed that neutral excitation of a crystal by light is possible with an electron bound to a hole in a lattice, the exciton, which travels from one atom of the lattice to another in a particle-like fashion.

An extension to the exciton theory can be made regarding molecular systems. When a molecule absorbs a photon that corresponds to a transition between two MO, the resulting electronic excited state is properly described by an exciton. In an early work devoted to the molecular exciton theory (MET), Simpson^{14a} represented the double bonds as quasi-independent weakly interacting systems. The excited states of polyene systems are described as an n -fold degenerate manifold, where n is the number of double bonds. Molecular excitons are categorized in two basic types: intrachain and interchain. The former are described by its movement along the π -conjugated polymer backbone, whereas the latter are formed when the separation between two molecular segments is small enough to allow a weak coupling (e.g., solid films, foldamers, and aggregates). Thus, the optical properties of the latter systems depend on the aggregation state of the polymer.³⁴

Another interesting case is the use of Frenkel exciton theory to study interference in the π -electron conjugation length of π -CPs. More specifically, Chang et al.^{14b} assumed that the MEH–PPV (i.e., a p -phenylene vinylene derivative) consists of a set of short π -COs with different lengths. On the basis of the conformational disorder caused by the torsional motion, the energy level of the emissive excited state of the oligomer was calculated using MET and assuming a single exciton localized in a finite chain. It is generally accepted that these phenylene-based conjugated systems acquire a planar configuration. However, occurrence of random distortions in the torsion

angles between adjacent monomer units could interfere with the behavior of the π -electrons, so that the observed conjugation length is much shorter than the physical polymer length.^{14b} Following the previous theory, Beenken et al.^{14c} studied the possible mechanism of segmentation of PTh into spectroscopic units (e.g., hydrogenated double bonds, kinks, and torsions) by using computer based molecular modeling methodologies.

Similarly, Bednarz et al.³⁵ studied the size-dependent relationship between absorption spectra of linear and cyclic Th-based oligomers. Their approach was based on a Frenkel exciton Hamiltonian, with an energy gap given by the equation

$$\epsilon_g^k = \omega_0 + 2J \cos\left(\frac{\pi k}{N+1}\right) \quad (14)$$

where $k = 1, 2, \dots, N$, ω_0 is the excitation energy of each repeating unit, and J describes the nearest-neighbor coupling.

More recently, Bednarz et al.³⁶ suggested the use of fitting functions derived from microscopic models based on the Frenkel exciton model. Hence, starting from the Pariser–Parr–Pople Hamiltonian to derive the excitonic form of an effective Hamiltonian based on a local description of the polymer in terms of monomers³⁷ and considering DFT results on Th- and phenylene-containing³⁹ π -COs, that the own authors demonstrated that excitations can be treated in the Frenkel exciton limit.³⁶ A lineal dependence of the oscillator strength (f) of the HOMO–LUMO transition was proposed for the larger Th-oligomers:

$$f = f_\infty - \frac{\text{const}}{N} \quad (15)$$

where f_∞ is the oscillator strength of the infinite polymer. Bednarz et al.³⁶ also provides the equations for the excitation energy of a model oligomer with two different blocks per repeating unit.

$$\begin{aligned} \epsilon_g^k &= \Delta \pm \sqrt{\delta^2 + 2J^2 + 2J^2 \cos\left(\frac{2k\pi}{N+1}\right)} \\ &= \Delta \pm \sqrt{\delta^2 + 4J^2 \cos^2\left(\frac{k\pi}{N+1}\right)} \end{aligned} \quad (16)$$

where $\Delta = (\omega_1 + \omega_2)/2$, $\delta = (\omega_1 - \omega_2)/2$, ω_1 and ω_2 are the on-site excitation energies of the two different repeating units within the unit cell, and J is the only interaction parameter between repeating units. Application of the excitonic approaches displayed in eqs 14 and 16 successfully described the experimentally observed size-dependent red shift of the optical spectra in Th- and phenylene-based π -COs series, results provided by the two approaches being very similar.

Beside their similarities, it is also interesting to compare eqs 8, 12, and 16, which describe HOMO–LUMO transitions of π -CPs using three very different theories (i.e., coupled oscillator, the HMO, and the Frenkel exciton, respectively).

Fitting Functions. In addition to the models proposed to describe HOMO–LUMO transition energy for different π -COs series, a set of fitting functions have also been used to adjust experimental and theoretical data (e.g., the linear and polynomial plots of ϵ_g against the inverse of the molecular length). In this scenario, other authors have introduced empirical fitting functions, some of which will be described in the following.

Since the seminal work of Lewis and Calvin,²² many empirical and theoretical formulas have been proposed to define the relationships between the structure of organic molecules, most of them involving conjugated chromophoric groups, and their UV–vis absorption spectra. The very simple straight line relationship between the square of wavelength and the conjugation length (eq 7) inspired the work of Hirayama,⁴⁰ which assumed the coupling of N vibrating chromophores in a conjugated system. Accordingly, the whole system was considered to vibrate as a simple harmonic oscillator with the restoring force of the conjugated chromophore

$$(\lambda_{\text{max}})^2 = A - BC^{N'} \quad (17)$$

where A , B , and C are empirical parameters to be adjusted, and N' is the homoconjugation index. The latter is the number of repeating units modified by a fraction to take into account the effect of the auxochromic substituents (i.e., in absence of such substituents, N' becomes the total repeating units $N' = n$).

In order to reduce the number of parameters in the Hirayama equation (eq 17), Meier et al.¹⁵ proposed a new exponential fitting function that accurately describes the absorption and fluorescence experimental data of the dialkoxy-substituted 1,4-phenylene-ethynylene π -COs series

$$\epsilon_g = \epsilon_{g,\infty} + (\epsilon_{g,1} - \epsilon_{g,\infty})e^{-a(n-1)} \quad (18)$$

where $\epsilon_{g,1}$ belongs to the parent compound ($n = 1$) and $\epsilon_{g,\infty}$ is the limiting value for $n \rightarrow \infty$. The parameter a indicates how fast the limit of convergence is approached.

When substituents are attached to the molecular backbone of π -COs, a significant shift on the adiabatic electronic transition can be induced by lowering their energies.⁴¹ When the substituents are exclusively attached to the end-units of the π -COs, the impact of the substituents is only manifested on the smaller oligomers. Gierschner et al.¹¹ proposed the use of a combination of the coupled oscillator model (eq 8) and an exponential term since the W. Kuhn expression cannot account for the push–pull-effect (i.e., it neglects the role played by intramolecular charge-transfer processes)

$$\epsilon_g = \epsilon_g^0 \sqrt{1 + 2 \frac{k'}{k_0} \cos\left(\frac{\pi}{N+1}\right)} - A e^{-bN} \quad (19)$$

where ϵ_g^0 , k'/k_0 , A , and b are the four parameters to be adjusted with the experimental or theoretical data. The influence of the exponential term is expected to decrease when the number of repeating units increases to the limit dominated by W. Kuhn's term.

As elaborated above, the seminal studies on the saturation of the electronic properties on π -CPs were devoted to the second hyperpolarizability, γ , as a function of chain length. Lu et al.⁴² proposed a four parameter fitting function to adjust experimental and theoretical values of polyacetylene (PA) polymers. Later, Luo et al.¹² adopted Lu's equation (eq 20) to study the saturation behavior of the optical band gap and dynamic hyperpolarizabilities of five-membered heteroaromatic oligomers, (Th-, pyrrole-, and furane-based oligomers). The same fitting function was applied to adjust the ϵ_g , α , and γ per repeating unit

$$\epsilon_g = \epsilon_{g,\infty} \left(\frac{1 - e^{-An}}{1 - B e^{-Cn}} \right) \quad (20)$$

where $\epsilon_{g,\infty}$, A , B , and C are the four parameters to be adjusted.

Accelerators of Convergence. Extrapolation techniques on electronic properties of π -CPs based on sequence transformations⁴³ have been used to accelerate the convergence at the infinite chain limit. A sequence transformation Γ is a rule that transforms a given sequence $\{s_n\}_{n=0}^{\infty}$ of calculated electronic property into a new sequence $\{s_n'\}_{n=0}^{\infty}$ with a faster convergence at the polymer limit. In fact, Cioslowski had considered in a seminal paper the use of Padé approximants in the extrapolation of the energy per unit cell at the bulk limit, which works particularly well for small oligomers.⁴⁴ Later Cui et al. obtain total energies per repeat unit of π -CPs with about 0.1 kcal/mol of accuracy using finite difference energy accelerators.³⁰ Moreover, Cioslowski et al.⁴⁵ applied 40 different extrapolation schemes in order to determine the limit of cluster energies. They show that Wynn's rho algorithm (eq 22), Wynn's epsilon (eq 23), and Aitken's iterated Δ^2 algorithm are those that produce more reliable results. Practically, there are two main classes of converging sequences, i.e., linearly and logarithmically convergent sequences. Aitken's iterated Δ^2 and Wynn's epsilon algorithms are powerful accelerators for linear convergences, whereas Wynn's rho algorithm is a powerful accelerator for logarithmic convergence. Similarly, Dalskov et al.⁴⁶ using this mathematical approach perform stable and reliable extrapolations of polarizabilities to the polymer limit. A more extended review on those methods of convergence acceleration has been done by Weniger et al.⁴⁷

The Wynn's rho algorithm⁴⁸ requires a sequence $\{x_n\}_{n=0}^{\infty}$ of interpolation points, which have to satisfy

$$0 < x_0 < x_1 < \dots < x_m < x_{m+1} < \dots$$

$$\lim_{n \rightarrow \infty} x_n = \infty \quad (21)$$

where the sequence transformation is

$$\rho_{-1}^{(n)} = 0, \quad \rho_0^{(n)} = s_n, \quad n \in \mathbb{N}_0$$

$$\rho_{k+1}^{(n)} = \rho_{k-1}^{(n+1)} + \frac{x_{n+k+1} - x_n}{\rho_k^{(n+1)} - \rho_k^{(n)}}, \quad k, n \in \mathbb{N}_0 \quad (22)$$

However, the Wynn's epsilon algorithm⁴⁸ does not involve the sequence of interpolation points

$$\varepsilon_{-1}^{(n)} = 0, \quad \varepsilon_0^{(n)} = s_n, \quad n \in \mathbb{N}_0$$

$$\varepsilon_{k+1}^{(n)} = \varepsilon_{k-1}^{(n+1)} + \frac{1}{\varepsilon_k^{(n+1)} - \varepsilon_k^{(n)}}, \quad k, n \in \mathbb{N}_0 \quad (23)$$

In both cases, only the elements with even subscripts, i.e., $\rho_{2k}^{(n)}$ and $\varepsilon_{2k}^{(n)}$, provide approximations to the limit s of the sequence $\{s_n\}_{n=0}^{\infty}$

Computational Details. The electronic properties of oligomers formed by n repeating units of thiophene (n -Th), 3,4-ethylenedioxythiophene (n -EDOT) and 3,4-phenylenedioxythiophene (n -PhEDOT), with n ranging from 2 to 30, have been determined not only in the gas-phase but also in acetonitrile and water using a Self-Consistent Reaction-Field (SCRf) method. Molecular geometries of n -Th, n -EDOT, and n -PhEDOT, in both the neutral and cation radical states have been fully optimized in both gas-phase and solution, results are compared. The choice of these three thiophene-based oligomers is based on the interest that those compounds have on the scientific community and the extended work on the subject that the group of the authors has developed on the last years about those π -CPs.⁴⁹ Poly(3,4-ethylenedioxythiophene),

PEDOT, is one of the most successful π -CPs due to a combination of properties, i.e., moderate band gap, low oxidation potential, high conductivity, good optical transparency, and exceptional environmental stability.⁵⁰ Poly(3,4-phenylenedioxythiophene), PPhEDOT, was explicitly designed to improve the solubility and processability of PEDOT but retaining the good electrical, electrochemical, and structural properties of the latter.⁵¹ Finally, polythiophene (PTh) was selected because it is the parent compound of the thiophene-based oligomers family under study.

Initially, gas-phase DFT geometry optimizations of the neutral and cationic forms of the considered π -COs were performed, the all *anti*-conformation (i.e., all the inter-ring dihedral angles arranged at 180°) being used as starting geometry in all cases. Calculations were carried out using the Gaussian09 computer program,⁵² by combining the Becke's three-parameter hybrid functional (B3)⁵³ with the Lee, Yang, and Parr (LYP)⁵⁴ expression for the nonlocal correlation, and the 6-31G(d) basis set (B3LYP/6-31G(d)). Also, the hybrid PBE0⁵⁵ functional and the Grimme's functional (B97D),⁵⁶ the latter being the one that includes dispersion, were also considered in combination with the 6-31G(d) basis set, PBE0/6-31G(d), and B97D/6-31G(d), on some of the calculated systems. The restricted formalism was considered for calculations on neutral oligomers (closed-shell systems), whereas monocations were considered in the doublet electronic state with the unrestricted DFT formalisms UB3LYP. This methodology was selected based on previous results, which indicated a very satisfactory description of the molecular geometry and relative energy for the minimum energy conformation of heterocyclic oligomers.^{16b,c} Although it is known that the reliability of this procedure is an effect of the cancellation of different errors,^{16b,57} the B3LYP/6-31G(d) is a very accurate method for the prediction of the electronic properties of π -CPs.^{16b,57,58} A complete geometry optimization of n -Th, n -EDOT, and n -PhEDOT structures was calculated in gas-phase, on both neutral and monocationic states, with n ranging from 2 to 30, 30, and 24, respectively. No symmetry constraints were imposed during these geometry optimizations, but torsion angle between consecutive rings of n -Th oligomers were fixed at 180° .

The polarity of the medium was considered to calculate the electronic properties of π -COs by means of reoptimizing the previously obtained gas-phase structures using a SCRf model. Specifically, the polarizable continuum model (PCM) developed by Tomasi et al.⁵⁹ was used to describe acetonitrile and water as solvents. PCM calculations on neutral and monocationic states were performed at the B3LYP/6-31G(d) and UB3LYP/6-31G(d) levels, respectively, and using the dielectric constants of acetonitrile ($\varepsilon = 35.69$) and water ($\varepsilon = 78.4$). Geometry optimizations in acetonitrile solution involved oligomers with n ranging from 2 to 30 (n -Th), 16 (n -EDOT), and 14 (n -PhEDOT), while for optimizations in aqueous solution, the largest values of n were 24, 12, and 12, respectively. In all cases, a low spin contamination was obtained, independently of the medium, the largest overestimation being of 2.6%.

The ε_g was calculated as the difference between the HOMO and LUMO energies (i.e., $\varepsilon_g = \varepsilon_{\text{HOMO}} - \varepsilon_{\text{LUMO}}$). Levy and Nagy evidenced that ε_g can be rightly approximated using this procedure in DFT calculations.⁶⁰ The vertical first ionization potential (IP_v) was calculated using Koopman's theorem (i.e., $\text{IP}_v = -\varepsilon_{\text{HOMO}}$), which, according to Janak's theorem, can also

be applied to DFT calculations.⁶¹ The first adiabatic ionization potential (IP_a) was calculated as the energy difference between the optimized structures of the monocationic and the neutral oligomers states (i.e., $IP_a = E_+ - E_0$).

RESULTS AND DISCUSSION

Choice of the Theoretical Procedure. Has been shown that ϵ_g values depend strongly on the model chemistry applied, being dependence focalized on the extent to which the exact exchange is incorporated in the theory.⁶² Here, we consider different model chemistries in order to see how they can affect on the electronic saturation and its relationship with system geometry. Full geometry optimization of n -EDOT series were performed using three different theoretical levels: B3LYP/6-31G (d), PBE0/6-31G, and B97D/6-31G. The latter has been employed to determine the effect of long-distance dispersion forces. Previous studies⁶³ showed the importance of dispersion forces on the equilibrium geometry of large systems, such as those studied in this work. Comparison among the results provided by the three theoretical methods will be through the ϵ_g extrapolation to infinite chain. Also, the polarity of the medium will be considered on the ϵ_g extrapolation study comparing these functionals.

Figure 2 plots the variation of ϵ_g against $\ln(1/n)$ calculated in gas-phase, acetonitrile and aqueous solution for the n -EDOT oligomer series. The fitting and extrapolation data are shown as

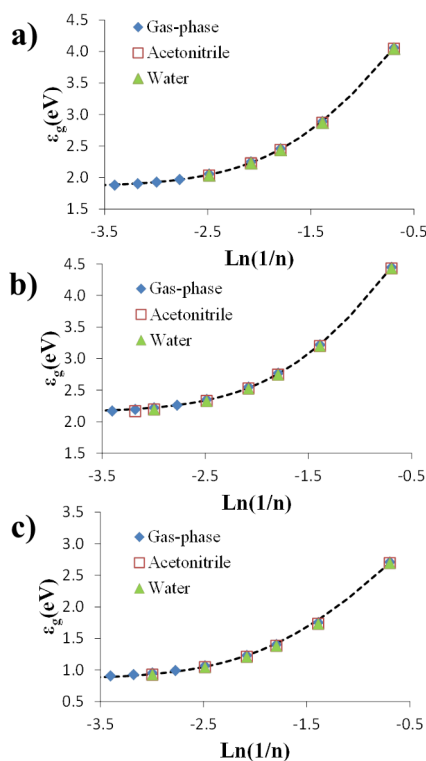


Figure 2. HOMO–LUMO gap (ϵ_g) against the natural logarithm of the reciprocal of the number of repeating units ($\ln 1/n$) calculated for n -EDOT π -conjugated oligomers at the (a) B3LYP/6-31G(d), (b) PBE0/6-31G(d), and (c) B97D/6-31G(d) levels in gas-phase (blue diamonds), acetonitrile solution (empty squares), and aqueous solution (green triangles). The implicit solvent was described using the PCM method. Extrapolations to infinite chain length (ϵ_g^∞) have been performed using the W. Kuhn fit (eq 8), which are represented by dashed lines.

a dashed line, the resulting values being very different depending on the functional used. Thus, the values extrapolated to infinite n applying the W. Kuhn model (eq 8) to theoretical data provided by the B3LYP functional are 1.85 and 1.83 eV in the gas-phase and solution (both acetonitrile and water), respectively. A small increase on the ϵ_g is obtained from extrapolated data using the PBE0 hybrid functional, being the ϵ_g value of 2.15 and 2.12 eV in the gas-phase and solution, respectively. However, when the dispersion effect is considered (B97D), we obtain a much lower extrapolated value of 0.85 and 0.81 eV in gas-phase and solution, respectively, which are away from expected experimental values of 1.6–2.0 eV.^{50a,64} In fact, there is a difference of about 1 eV between experimental and theoretical limit values when dispersion effects are considered using the B97D functional. The shorter values of ϵ_g obtained with this functional, can be explained due to the observed diminution on the aromaticity and BLA indices on calculated systems. It is known the relationship between aromaticity, BLA, and ϵ_g , being the latter more easy to control through the change on the aromaticity of π -CPs.⁶⁵ In order to clarify this difference on the tendency behavior of ϵ_g when long-range interactions are included, a short study of system's aromaticity is performed (see Supporting Information). It has been shown that electronic properties of π -COs have been related with their aromatic character.⁶⁶ Thus, comparing aromaticity indices (HOMA⁶⁷) of 4-EDOT and 12-EDOT at different polarity environments show an important reduction on the HOMA values, which becomes close to that of a kekulé structure when dispersion forces are considered. Indeed, an important reduction on the aromaticity index involves a diminution on the ϵ_g energy as seen in previous works.^{66a} Moreover, the influence of the polarity of the medium is very small on the extrapolation of ϵ_g .

Taken into account the best estimation of the electronic properties, such as the ϵ_g , and by consistency with previous calculations,^{16b,d} hereafter, all the calculations refer to those performed using the B3LYP functional only. However, a short comparison between different model chemistry will be conducted after considering model fitting.

Model Fitting. Band Gap. The lack of comprehensive studies comparing the different models proposed to extrapolate the electronic properties of π -COs to infinite chain length has motivated the current investigation. Thus, in order to complete the review work reported in the previous sections, here, we present a practical description of the goodness of the different models when applied to ϵ_g , IP_v , and IP_a of π -conjugated oligomers. Specifically, models have been adjusted to theoretical data obtained for n -Th, n -EDOT, and n -PhEDOT series. In order to examine the affects of the saturation on the electronic properties of each model, two different regressions were performed. In the first adjustment, the elements of each series with smaller n (i.e., $n = 2, 4, 6$, and 8) were the only used, whereas all the elements were considered in the second adjustment (i.e., the value of n extends from 2 to the maximum chain length of each series). Considering an ECL located at about 12 repeating units,¹⁵ the ϵ_g value extrapolated from the first and second adjustments corresponds to π -COs before and after the ECL point. However, the ECL parameter was found to change significantly with the chemical nature of the π -COs series.¹⁵ Gierschner et al.¹¹ proposed the use of the maximum conductive chain length (N_{MCC}) instead, which is similar to the ECL parameter but less sensitive to the chemical nature of the π -COs. The N_{MCC} parameter is obtained graphically from the

plot of ϵ_g^∞ versus $1/n$ as the intersection of the extrapolation of the linear part of the curve with the horizontal line representing the polymer limit. The N_{MCC} parameters obtained for the calculated n -Th, n -EDOT, and n -PhEDOT π -COs series in this work are 20.5, 25.3, and 20.8, respectively. Consequently, this will allow us to determine the deviation of the model due to the effect of the saturation of the electronic property in the polymer by comparing both shorter and longer oligomer series. Specifically, a small deviation is expected from those models that show a good fit on the saturation effect.

Table 1 lists the results derived from fitting the theoretical data calculated for n -Th to all previously discussed models.

Table 1. Comparison between the Different Models Discussed in This Work^a

model	eq	$n \leq 8$		$n = 2-30$		$\Delta_{eg}(\%)$
		ϵ_g^∞	R^2	ϵ_g^∞	R^2	
Hirayama	17	2.08	0.999984	2.09	0.999916	0.4
Lewis and Calvin	7	1.95	0.998369	1.94	0.998759	0.5
W. Kuhn	8	2.02	0.999636	2.05	0.999667	1.1
HMO mod.	12	2.10	0.999994	2.06	0.999914	1.7
MET (2mon)	16	2.12	0.999973	2.07	0.999708	2.4
FEMO	2	2.06	0.995782	1.99	0.995762	3.3
linear (1/N)	3a	1.81	0.999935	1.88	0.997742	3.6
MET (1mon)	14	2.21	0.998107	2.11	0.993866	4.3
Gierschner	19	2.16	1.000000	2.06	0.999969	4.7
Lu	20	2.21	1.000000	2.10	0.999721	4.9
HMO	10	1.71	0.999724	1.84	0.994116	7.4
FEMO ext.	5	1.79	0.999972	1.92	0.999908	7.6
quadratic (1/N)	3b	1.79	0.999974	1.92	0.999090	7.6
Meier	18	2.33	0.999722	2.14	0.995332	8.1
Wynn's rho ^b	22			2.01 $\pm$ 0.01		
theory (eV)				2.03 ^c		

^aExtrapolation of the theoretical data obtained for n -Th oligomers has been used to get the energy band gap (ϵ_g^∞ , eV) at the polymer limit (PTh). Calculations were performed at the B3LYP/6-31G(d) level. The differences between the band gaps derived from oligomers with $n \leq 8$ and from all the calculated oligomers (n from 2 to 30) are indicated (Δ_{eg} , %). The regression coefficient (R^2) and the fitting model with its equation number (eq) are also shown. ^b n -Th oligomers with $n = 4, 8, 12, 16, 20$, and 24. ^cValue obtained for 50-Th (ref 16b).

More specifically, it includes the ϵ_g^∞ values, the correlation coefficients, and the difference in percentile between the ϵ_g^∞ values extrapolated for a given model using the results for oligomers with $n \leq 8$ and for all calculated oligomers (Δ_{eg}).

Parameters displayed in Table 1 are ranked according to Δ_{eg} , the three-parameters model of Hirayama (eq 17) being the one with the lowest difference between the two extrapolations. Indeed, saturation effect on the band gap at infinite chain fits very well for both Hirayama ($\Delta_{eg} = 0.4\%$) and Lewis and Calvin ($\Delta_{eg} = 0.5\%$) models, even though the correlation coefficient of the latter is slightly worse. Interestingly to notice from the bibliography, the theoretical value of the extrapolated PTh band gap ($\epsilon_g^\infty = 2.03$ eV)^{16b} is obtained at the same level of calculations as the current work but with using larger oligomers up to 50-Th. Considering the latter as a reference point and comparing with the tabulated values, the coupled oscillator model of W. Kuhn (eq 8) presents a very close predicted value even in the case of just four points adjustment ($\epsilon_g^\infty = 2.02$ eV). The modified HMO model (eq 12) also provides a satisfactory

prediction of the polymer band gap using both a shorter and longer number of fitting points.

Although all the fitted models present a reasonably good adjustment with regression coefficients higher than 0.999, the extrapolation leads some to values that differ significantly from the expected one ($\epsilon_g^\infty = 2.03$ eV). For instance, the regression coefficients of both the extended FEMO (eq 5) and the quadratic polynomial models (eq 3b) are very good (>0.9999), even though the extrapolated values are considerably smaller than the expected one.

Table 2 lists the results derived from the different models for n -EDOT oligomers. In this case, the coupled oscillator model

Table 2. Comparison between the Different Models Discussed in This Work^a

model	eq	$n \leq 8$		$n = 2-30$		$\Delta_{eg}(\%)$
		ϵ_g^∞	R^2	ϵ_g^∞	R^2	
W. Kuhn	8	1.86	0.999905	1.85	0.999922	0.2
Gierschner	19	1.90	1.000000	1.87	0.999908	1.3
linear (1/N)	3a	1.66	0.999556	1.69	0.999024	1.6
HMO mod.	12	1.90	0.999991	1.85	0.999881	2.2
MET (2mon)	16	1.93	0.999954	1.88	0.993374	2.6
Lewis and Calvin	7	1.80	0.996714	1.75	0.997294	2.7
Hirayama	17	1.83	0.999978	1.88	0.999924	2.8
FEMO	2	1.90	0.993726	1.80	0.992468	5.4
Lu	20	2.00	1.000000	1.90	0.999697	5.4
HMO	10	1.56	0.999979	1.65	0.997059	5.4
MET (1mon)	14	2.05	0.996642	1.93	0.989400	6.1
quadratic (1/N)	3b	1.57	0.999984	1.70	0.999168	8.4
FEMO ext.	5	1.56	0.999980	1.70	0.999177	8.8
Meier	18	2.14	0.999721	1.94	0.995179	9.5
Wynn's rho ^b	22			1.79 $\pm$ 0.01		
theory (eV)				1.80 ^c , 1.86 ^d		

^aExtrapolation of the theoretical data obtained for n -EDOT oligomers has been used to get the energy band gap (ϵ_g^∞ , eV) at the polymer limit (PEDOT). Calculations were performed at the B3LYP/6-31G(d) level. The differences between the band gaps derived from oligomers with $n \leq 8$ and from all the calculated oligomers (n from 2 to 30) are indicated (Δ_{eg} , %). The regression coefficient (R^2) and the fitting model with its equation number (eq) are also shown. ^b n -EDOT oligomers with $n = 4, 8, 12, 16, 20$, and 24. ^cValue obtained for 30-EDOT (ref 16b). ^dValue obtained for 16-EDOT (ref 16d).

of W. Kuhn is the one with a smaller difference between extrapolated polymer band gap values ($\Delta_{eg} = 0.2\%$). It is worth noting that the correlation coefficient of this model does not decrease when more fitting points closer to the zone of saturation of the system are included, contrary to what happens in many of the tested models. However, in this system, both the Hirayama and modified HMO models present some major differences in their Δ_{eg} values, being more affected by the saturation of ϵ_g with the chain length. However, the ϵ_g^∞ values extrapolated for such three models are very close among them as well as to other theoretical values reported in the literature, which were obtained using similar levels of calculation. Furthermore, in good agreement with the bibliography,¹⁵ usual ϵ_g extrapolation as a function of $1/n$ presents defect estimation on both linear and quadratic fitting models. This is more evident on the linear fit than in a quadratic fit, being this difference of about 0.1 eV compared with the expected values. Experimental ϵ_g^∞ of the polymer limit has been determined in the range of 1.6 up to 1.9 eV,^{50a,64a} being the PEDOT film ϵ_g

estimated recently as 1.98 eV from electrochemical data on a previous work of the authors.^{64b}

The fitting models describing ϵ_g evolution with the chain length of PPhEDOT are shown in Table 3. In these π -COs, the

Table 3. Comparison between the Different Models Discussed in This Work^a

model	eq	$n \leq 8$		$n = 2-24$		$\Delta_{eg}(\%)$
		ϵ_g^∞	R^2	ϵ_g^∞	R^2	
W. Kuhn	8	2.16	0.999983	2.15	0.999969	0.2
Hirayama	17	2.18	0.999987	2.19	0.999920	0.4
HMO mod.	12	2.17	0.999996	2.15	0.999954	0.8
Lewis and Calvin	7	2.05	0.997713	2.03	0.998162	1.1
MET (2mon)	16	2.19	0.999972	2.22	0.993323	1.2
linear (1/N)	3a	1.94	0.999691	1.96	0.998806	1.2
FEMO	2	2.16	0.994323	2.09	0.993794	3.1
MET (1mon)	14	2.29	0.997094	2.22	0.993323	3.2
Gierschner	19	2.25	1.000000	2.17	0.999914	3.2
HMO	10	1.85	0.999939	1.92	0.996815	3.5
Lu	20	2.28	1.000000	2.19	0.999744	3.9
Meier	18	2.38	0.999757	2.25	0.995967	5.3
FEMO ext.	5	1.89	0.999972	2.00	0.999179	5.8
quadratic (1/N)	3b	1.88	0.999970	1.99	0.999032	6.0
Wynn's epsilon ^b	23			2.16 ± 0.01		
theory (eV)				2.24^c		

^aExtrapolation of the theoretical data obtained for n -PhEDOT oligomers has been used to get the energy band gap (ϵ_g^∞ , eV) at the polymer limit (PPhEDOT). Calculations were performed at the B3LYP/6-31G(d) level. The differences between the band gaps derived from oligomers with $n \leq 8$ and from all the calculated oligomers (n from 2 to 24) are indicated (Δ_{eg} , %). The regression coefficient (R^2) and the fitting model with its equation number (eq) are also shown. ^b n -PhEDOT oligomers with $n = 4, 8, 12, 16, 20$, and 24. ^cValue obtained for 14-PhEDOT (ref 16d).

coupled oscillator of W. Kuhn, the Hirayama fitting function, and the modified HMO have the lower Δ_{eg} (<1%) and the higher correlation coefficient in both correlations, before and after N_{MCC} parameter. Their extrapolated ϵ_g^∞ values are close to the expected value but much higher than experimental available datum, which has been attributed to the level of calculation.

The average values of Δ_{eg} , which have been calculated using the parameters obtained for the three studied π -CPs, are listed in Table 4, the tabulated models being ranked by the resulting Δ_{eg} average. As seen on previous systems, the models with the lowest difference Δ_{eg} are the coupled oscillator W. Kuhn ($\Delta_{eg} = 0.5\%$) and the Hirayama fitting function ($\Delta_{eg} = 1.2\%$). Interestingly, a model as simple as that of Lewis and Calvin also has a rather small difference ($\Delta_{eg} = 1.4\%$). However, as discussed above, this model leads to a ϵ_g^∞ value that differs from the expected value compared to the previous two. Also, an interesting measure is the difference between experiment and the best extrapolation models. Table 5 lists the percentage of deviation of the ϵ_g^∞ with respect to the midpoint on experimental range values, E_{exp} . As can be seen, a very similar dispersion between the ϵ_g^∞ and experimental values was obtained between the four models fitting with smaller Δ_{eg} , the averaged dispersion being of $E_{exp} = 12.6\%$. Similar behaviors of these models fitting are expected on other π -CPs since these methods had been applied to a broad variety of conjugated polymers, as has been shown above. Also, the deviation on

Table 4. Average Band Gap Difference between the Fittings Obtained Using Oligomers with $n \leq 8$ and All the Calculated Oligomers (Δ_{eg} , %)^a

model	eq	$\Delta_{eg}(\%)$
W. Kuhn	8	0.5
Hirayama	17	1.2
Lewis and Calvin	7	1.4
HMO mod.	12	1.6
MET (2mon)	16	2.1
linear (1/N)	3a	2.1
Gierschner	19	3.1
FEMO	2	3.9
MET (1mon)	14	4.5
Lu	20	4.7
HMO	10	5.4
quadratic (1/N)	3b	7.4
FEMO ext.	5	7.4
Meier	18	7.6
		3.8 ^b

^aAverages have been performed using the results obtained for n -Th, n -EDOT, and n -PhEDOT. The fitting model with its equation number (eq) is shown. ^bColumn average.

Table 5. Comparison between Experimental and the Best Extrapolation Models^a

model	eq	n -Th		n -EDOT		n -PhEDOT	
		ϵ_g^∞	E_{exp}	ϵ_g^∞	E_{exp}	ϵ_g^∞	E_{exp}
W. Kuhn	8	2.05	4.7	1.85	2.8	2.15	30.3
Hirayama	17	2.09	2.8	1.88	4.4	2.19	32.7
Lewis and Calvin	7	1.94	9.8	1.75	2.8	2.03	23.0
HMO mod.	12	2.06	4.2	1.85	2.8	2.15	30.3
exptl (eV)		$2.1-2.2^b$		$1.6-2.0^c$		1.65^d	

^aThe extrapolation of the theoretical data obtained for n -Th, n -EDOT, and n -PhEDOT oligomers (with n up to 30, 30, and 24, respectively) has been used to get the energy band gap (ϵ_g^∞ , eV) at the polymer limit. Calculations were performed at the B3LYP/6-31G(d) level. The error comparing the given theoretical extrapolation with mid value of experimental data range (E_{exp} , %) and the fitting model with its equation number (eq) are also shown. ^bFrom ref 70. ^cFrom refs 50a and 64. ^dFrom ref 51b.

deriving properties at the polymer limit, in some of the extrapolation methods reported in the bibliography using experimental data, is also reproduced.¹¹

Finally, sequence transformation (eqs 22 and 23) has been applied to the ϵ_g values derived from the oligomer series made of $n = 4, 8, 16, 20$, and 24 repetitive units on the three calculated systems as accelerator of convergence, being obtained the ϵ_g^∞ values of 2.01 ± 0.01 , 1.79 ± 0.01 , and 2.16 ± 0.01 for the n -Th, n -EDOT (Wynn's rho algorithm), and n -PhEDOT systems (Wynn's epsilon algorithm), respectively (Tables 1–3). The asymptotic condition $\lim_{n \rightarrow \infty} (s_{n+1} - s)/(s_n - s) = \rho$, proposed on the bibliography⁴⁷ to discern between linearly ($|\rho| < 1$) and logarithmically ($\rho = 1$) convergences has been used. The ϵ_g values of n -PhEDOT oligomer series present a closer linearly convergence. Thus, n -Th and n -EDOT oligomers series are extrapolated using Wynn's rho algorithm, whereas n -PhEDOT is extrapolated using Wynn's epsilon algorithm. We want to notice that the average error comparing the shorter extrapolation using only four oligomers in the series ($n = 2$ up to 8) with the sequence transformation are $\sim 1.5\%$, $\sim 2.2\%$, and $\sim 9\%$ for the W. Kuhn,

Hirayama, and linear $1/n$ extrapolation models, respectively. That will allow to perform a good estimation of the polymer limit using only the first elements of an oligomer series by using the simple W. Kuhn's model.

Ionization Potential. In a previous work,^{16d} it was shown that polarity of the medium has virtually no effect on the theoretical ε_g data as a function of the chain length. This is quite the opposite of what happens to the ionization potential, which shows an appreciable difference when values obtained in the gas-phase and in solution are compared. The most used models (i.e., linear and quadratic fits with the inverse of the chain length) have been compared with the Hückel model (HMO) to evaluate the eigenvalue orbital energy (eq 9) since the IP_v can be approximated as the $-E_{HOMO}$ following Koopman's theorem.⁶⁸

Table 6 lists the extrapolated values of IP_v for the three studied systems considering different fitting models. The

Table 6. Comparison between Different Extrapolation Models Used to Obtain the Vertical Ionization Potential (IP_v^∞ , eV) at the Infinite Chain Length of the Three Series of Calculated Oligomers: i.e., n -Th, n -EDOT, and n -PhEDOT; Correlation Coefficients Are Also Shown (R^2)

	gas-phase		acetonitrile		water	
	IP_v	R^2	IP_v	R^2	IP_v	R^2
PTh						
linear (1/N)	4.52	0.9914	4.58	0.9991	4.58	0.9991
quadratic (1/N)	4.56	0.9986	4.59	0.9996	4.60	0.9996
HMO	4.61	0.9990	4.70	0.9950	4.70	0.9950
$IP^{\infty a}$	4.61		4.66		4.66	
exptl IP			4.9 ^b			
PEDOT						
linear (1/N)	3.44	0.9994	3.93	0.9996	3.94	0.9996
quadratic (1/N)	3.43	0.9996	3.90	0.9999	3.92	0.9999
HMO	3.60	0.9854	4.10	0.9939	4.11	0.9940
$IP^{\infty a}$	3.53		4.05		4.04	
exptl IP			4.1–4.3 ^c			
PPhEDOT						
linear (1/N)	4.28	0.9995	4.52	0.9994	4.53	0.9993
quadratic (1/N)	4.28	0.9995	4.51	0.9996	4.51	0.9997
HMO	4.42	0.9942	4.64	0.9924	4.66	0.9932
$IP^{\infty a}$	4.38		4.60		4.61	
exptl IP			4.6 ^d			

^aLimit extrapolation using a high order polynomial. ^bFrom ref 71.

^cFrom refs 51a, 64b, and 69. ^dFrom ref 69.

quadratic and linear fits present a better correlation coefficient than the HMO model. Moreover, the latter has a minor error of extrapolation mainly due to its proximity to the asymptotic limit of infinite chain axis abscises, where $\cos[\pi/(n+1)] = 1$. In fact, comparison of extrapolated values with the ones obtained through a polynomial fit of high degree as a function of the cosine reveals a larger distance to the theoretical limit when using both linear and quadratic methods than with the simple HMO. Specifically, inspection of the IP_v extrapolation limits (IP_v^∞) reveals similar effects due to the polarity medium for the three oligomer series r . The difference between the IP_v^∞ calculated in the gas-phase and in solution is 0.05, 0.51–0.52, and 0.22–0.23 eV for the PTh, PEDOT, and PPhEDOT,

respectively. It is worth noting the IP experimental values of PEDOT derived from electrochemical data that were estimated as 4.1–4.3 eV,^{51a,64b,69} close to the IP_v^∞ value of this work (4.05 eV, Table 6) in acetonitrile.

Table 7 lists the extrapolated values of IP_a . As was found for IP_v , the extrapolation using the HMO model gives values very

Table 7. Comparison between Different Extrapolation Models Used to Obtain the Adiabatic Ionization Potential (IP_a^∞ , eV) at the Infinite Chain Length of the Three Series of Calculated Oligomers: i.e., n -Th, n -EDOT, and n -PhEDOT; Correlation Coefficient Is Also Shown (R^2)

	gas-phase		acetonitrile		water	
	IP_a	R^2	IP_a	R^2	IP_a	R^2
PTh						
linear (1/N)	4.90	0.9914	4.36	0.9951	4.38	0.9890
quadratic (1/N)	4.79	0.9996	4.43	0.9992	4.46	0.9985
HMO	5.16	0.9631	4.50	0.9999	4.50	0.9994
$IP^{\infty a}$	4.94		4.53		4.53	
PEDOT						
linear (1/N)	3.81	0.9920	3.71	0.9984	3.71	0.9981
quadratic (1/N)	3.67	1.0000	3.75	0.9995	3.76	0.9994
HMO	4.09	0.9617	3.88	0.9991	3.88	0.9993
$IP^{\infty a}$	3.84		3.88		3.86	
PPhEDOT						
linear (1/N)	4.69	0.9960	4.31	0.9953	4.31	0.9953
quadratic (1/N)	4.58	0.9999	4.37	0.9990	4.37	0.9990
HMO	4.94	0.9816	4.44	0.9997	4.43	0.9997
$IP^{\infty a}$	4.74		4.46		4.46	

^aLimit extrapolation using a high order polynomial.

close to the extrapolated theoretical limit (i.e., data fit by a polynomial of high degree as a function of the cosine). Therefore, the use of the HMO model is recommended to extrapolate the IP 's. However, if a greater accuracy in the extrapolation is required, the combination of the HMO model with higher polynomial degrees as a fitting function, instead of the linear one, is suggested.

Environment Polarity Effect. Molecular Geometry. The influence of the polarity of the medium in the molecular geometry of 12-Th, 12-EDOT, and 12-PhEDOT has been examined by comparing bond distances and angles in both neutral and oxidized states in a growing medium polarity (i.e., gas-phase, acetonitrile, and water). It should be mentioned that the oligomers with $n = 12$ were selected because they are located around the midpoint of the observed saturation effect of electronic properties (see previous sections). The oxidized state was considered because the local distortion to form a quinoid-type structure on the central region may be affected by the environment.⁷

A study on the differences between the bond distances and angles (Δd_b and $\Delta \theta_b$, respectively) for the central repeating unit on the 12-Th and 12-Th⁺ oligomers reflects that both bond distances and angles on the neutral system do not show significant variations when the molecular geometry is optimized in different environments (Supporting Information, Table ST3). In opposition, the oxidized state presents appreciable differences. Positive and negative variations of bond distances with respect to the neutral state are leading to a quinoid-type structure of the Th ring. Moreover, geometry optimizations with implicit solvents stress even more the quinoid-type

structure, even though there is no difference between the structures obtained in acetonitrile and water.

The differences Δd_b and $\Delta \theta_b$ obtained for 12-EDOT and 12-EDOT⁺ show similar behavior to the Th-oligomer, where the formation of the quinoid-type structure in the oxidized state is more evident when a polar environment is considered (Supporting Information, Table ST4). Geometric parameters involved in the ethylenedioxy bridge are also affected by the polarity of the environment, even though there is no significant difference between acetonitrile and water.

Differences in Δd_b and $\Delta \theta_b$ are listed on Table ST5, Supporting Information, for 12-PhEDOT and 12-PhEDOT⁺. Retrospectively, we compare with previous Δd_b oligomer differences (i.e., the oligomer series 12-Th, 12-EDOT, and 12-PhEDOT) between oxidized and neutral states. It is worth noting that Δd_b values are becoming slightly larger in gas-phase; whereas, the largest differences in a polar environment become shorter as we decrease the molecular dipole moment. Regarding bond angles differences ($\Delta \theta_b$), no appreciable differences are observed beyond those due to replace the two carbon atoms from the ethylenedioxy bridge by a benzene ring.

A general view on the two possible limit structures adopted by the studied oligomers (benzenoid- and quinoid-type structures), are represented in Figures SF4 and SF5, Supporting Information. These figures represent the C–C bond distances along the three oligomer backbones (i.e., 12-Th, 12-EDOT, and 12-PhEDOT) in their neutral and oxidized states, respectively. As can be seen in Figure SF2 (Supporting Information), the structure of the neutral state is practically independent of the polarity of the environment. Thus, the bond length alternation patterns are similar in all of them. However, a very different behavior is observed for the backbone C–C bond distances of the oligomers in the oxidized state (Figure SF3, Supporting Information). The charged oligomer has a clear quinoid-type structure on the molecular central region where the consecutive C–C bond distances are getting close to each other and with a clear decrease of bond distances between consecutive thiophene rings. Indeed, this kind of structure is getting diluted as we move away from the center of the oligomer molecule. In addition, it is worthy to notice that bond distance alternation in the central region depends on the polarity of the medium. For example, bond lengths obtained in the water for the central regions of 12-Th⁺ (Figure SF3a, Supporting Information) and 12-PhEDOT⁺ (Figure SF3c, Supporting Information) behave completely different with respect to the gas-phase. This coincides with what has already been shown in Tables ST3, ST4, and ST5, Supporting Information. The polarity of the solvent assists in the stabilization of the quinoid-type structures, this effect increasing with the dipole moment of the oxidized molecule.

Charge and Spin Distribution. The Mulliken population analysis was used to calculate the distribution of charges along the repeating units of 12-Th⁺, 12-EDOT⁺, and 12-PhEDOT⁺ oligomers (Figure 3). Independently of the environment, charges move from the end-chain to the middle. However, there is a pronounced tendency to concentrate charge in the central region when a polar environment is considered. Thus, gas-phase calculations provide an accumulation of charge at the center: 25, 30, and 29% of charge accumulation in the four central monomers of the 12-Th⁺, 12-EDOT⁺, and 12-PhEDOT⁺, respectively. This behavior is magnified when implicit aqueous solvent is considered, the charge accumulated

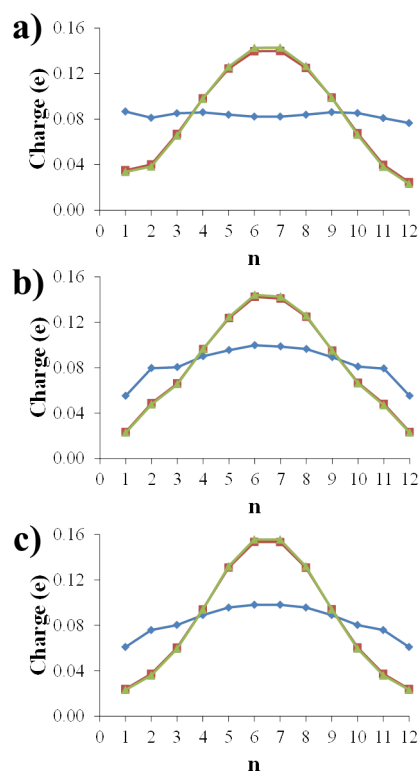


Figure 3. Mulliken charges per repeating unit calculated for oxidized oligomers: (a) 12-Th⁺, (b) 12-EDOT⁺, and (c) 12-PhEDOT⁺. Three environmental conditions are shown: gas-phase (blue diamonds), acetonitrile solution (brown squares), and aqueous solution (green triangles).

in the four central repeating units of the 12-Th⁺, 12-EDOT⁺, and 12-PhEDOT⁺ being 41, 41, and 44%, respectively.

However, to clarify the charge transport mechanism, we also evaluated the spin density per repeating unit in the same oxidized oligomers (Figure 4). As it was found for the charge, the spin density moved from the end-chain to the center. Furthermore, the distribution of both charges and spin densities are similar with exception of the gas-phase, some differences being detected in this environment. The largest concentration of spin density is observed on those systems calculated in solution. In fact, in the gas-phase, the spin density accumulation on the four central monomers are 25, 35, and 34% for 12-Th⁺, 12-EDOT⁺, and 12-PhEDOT⁺, respectively, increasing to 44, 44 and 48% in aqueous solution.

CONCLUSIONS

In this work, we have reviewed different strategies to extract the π – π^* lowest transition energy from the polymer limit. Different fitting models and fitting functions considering saturation effects of electronic properties at growing chain length have been examined from a practical point of view. The ϵ_g values derived from three series of Th-based oligomers, i.e., *n*-Th, *n*-EDOT, and *n*-PhEDOT with up to 30, 30, and 24 monomers, respectively, were used to fit the ϵ_g for the corresponding π -CPs and compare different strategies and extrapolation procedures. Moreover, the effects of the polarity of the environment on the three series of π -COs, in both neutral and oxidized states, have been considered. The following conclusions have been drawn:

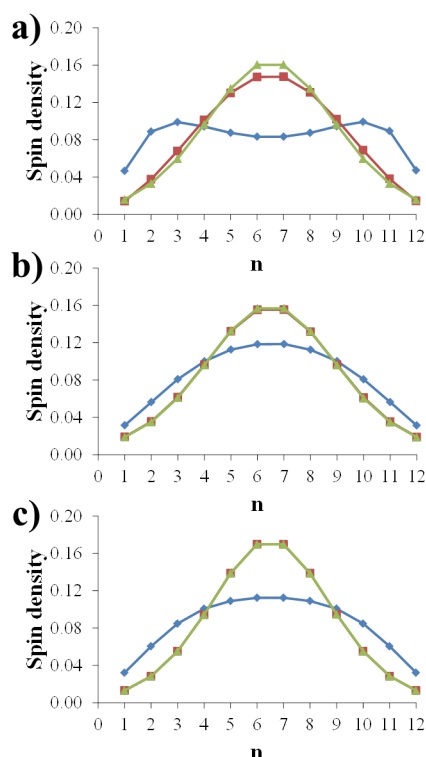


Figure 4. Spin density per repeating unit calculated for oxidized oligomers: (a) 12-Th⁺, (b) 12-EDOT⁺, and (c) 12-PhEDOT⁺. Three environmental conditions are shown: gas-phase (blue diamonds), acetonitrile solution (brown squares), and aqueous solution (green triangles).

- (i) The coupled oscillator model of W. Kuhn (holding two fitting parameters) provides the best results when the ϵ_g is adjusted as a function of chain length. This model presents the lowest dispersion of ϵ_g^∞ values allowing to extrapolate the transition energy on the polymer limit from the first elements of the oligomer series. It has been shown that only four oligomers of the series ($n = 2$ up to 8) allow an extrapolation value of ϵ_g^∞ with a $\sim 1.5\%$ of average error.
- (ii) The three-parameter fitting function of Hirayama leads to satisfactory results with both their ϵ_g interpolation and extrapolation to the polymer limit from the first elements of the series. However, the more extended fitting method by means of linear and quadratic polynomial functions against the inverse of the chain length are leading to more disperse values on the polymer limit, making necessary a larger number of elements from the oligomer series to obtain reasonable results.
- (iii) Considering saturation effects on the ionization potential of the three π -COs series has shown that the HMO model is able to extrapolate theoretical values of IP_v and IP_a to the polymer limit with low errors. Comparison of this model with the linear and quadratic polynomial fits indicates that the better fit of the former is due to an observed asymptote at infinite chain length, which allows an extrapolation with a lower error on the polymer infinite.
- (iv) The solvent polarity helps to stabilize the quinoid-type structure, this stabilization growing with the dipole moment of the oxidized molecule. The π -CO aromaticity increases with the polarity of the medium, favoring the

reduction of the alternation in the bond distance of the Th ring.

■ ASSOCIATED CONTENT

Supporting Information

Tables and figures giving a detailed description of the systems aromaticity and systems geometry of all involved π -CPs at different polarity environments. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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